Ray Tracing Attention

Output token

Vocabulary Tree
Token-space geometry

Ray Tracing Readout
LM Head geometry
Strongest signal → token

RMSNorm

RTA layers

High-D layer state
Restore · pass to next layer

Weight geometry
Feature Tree
Optical coefficients

Ray Tracing MLP
Gate / up light response
Down readout → high-D

N×

RMSNorm

Context geometry
History Tree
Recurrent light field

Ray Tracing Attention
Projection · RoPE · ray query
Detector → high-D signal

RMSNorm

Cached path

Cached token
Tree hit → token

HIT

MISS

Causal Token Tree
Q = full context path
One query per output position

Full-model path

Token embedding
Source model coefficients

Input context
Prompt + generated tokens